

1. Prob 16.1

The central potential is

$$V(r) = \frac{1}{2}m\omega_0^2 r^2 \quad (1)$$

Putting this into (16.13) gives

$$\frac{dE}{dt} = -\frac{\tau}{m} \langle (m\omega_0^2 r)^2 \rangle = -\tau\omega_0^2 \langle m\omega_0^2 r^2 \rangle \quad (2)$$

For a harmonic oscillator, we can identify $\langle m\omega_0^2 r^2 \rangle$ as the energy itself, giving

$$\frac{dE}{dt} = -\tau\omega_0^2 E = -\Gamma E \quad \Rightarrow \quad E = E_0 e^{-\Gamma t} \quad (3)$$

For angular momentum, (16.16) also gives

$$\left\langle \frac{d\mathbf{L}}{dt} \right\rangle = -\frac{\tau}{m} \left\langle \frac{1}{r} \frac{dV}{dr} \right\rangle \mathbf{L} = -\tau\omega_0^2 \mathbf{L} = -\Gamma \mathbf{L} \quad \Rightarrow \quad \mathbf{L} = \mathbf{L}_0 e^{-\Gamma t} \quad (4)$$

2. Quantum Mechanical Calculation

The following illustrates the quantum mechanical calculation of the energy and angular momentum decay of an oscillator.

With the conjugate momentum (12.14)

$$\mathbf{P} = \mathbf{p} + \frac{e}{c} \mathbf{A} \quad (5)$$

the total Hamiltonian of a charged particle in an electromagnetic field is given in (12.17)

$$\begin{aligned} H &= \sqrt{(c\mathbf{P} - e\mathbf{A})^2 + m^2 c^4} + e\Phi + H_{\text{field}} && \text{under nonrelativistic limit} \\ &\approx mc^2 + \frac{1}{2m} \left(\mathbf{P} - \frac{e}{c} \mathbf{A} \right)^2 + e\Phi + H_{\text{field}} \\ &= mc^2 + \frac{p^2}{2m} - \frac{e}{2mc} (\mathbf{A} \cdot \mathbf{P} + \mathbf{P} \cdot \mathbf{A}) + \frac{e^2 A^2}{2mc^2} + e\Phi + H_{\text{field}} && [\mathbf{P}, \mathbf{A}] = 0 \text{ for Coulomb gauge} \\ &= \underbrace{mc^2 + \frac{p^2}{2m} + e\Phi + H_{\text{field}}}_{H_0} - \frac{e}{mc} \mathbf{A} \cdot \mathbf{P} + \frac{e^2 A^2}{2mc^2} \\ &= H_0 - \frac{e}{mc} \mathbf{A} \cdot \mathbf{p} - \frac{e^2 A^2}{2mc^2} \end{aligned} \quad (6)$$

Since $A \sim a^\dagger + a$, A^2 changes photon number by 0 or ± 2 , thus is of second order compared to the interaction term $-(e/mc)\mathbf{A} \cdot \mathbf{p}$. We thus identify H_0 as the unperturbed Hamiltonian and

$$H' = -\frac{e}{mc} \mathbf{A} \cdot \mathbf{p} \quad (7)$$

as the perturbation Hamiltonian. By Fermi's Golden Rule, the transition rate from an initial state $|i\rangle$ to a final state $|f\rangle$ is given by

$$\Gamma_{i \rightarrow f} = \frac{2\pi}{\hbar} |\langle f | H' | i \rangle|^2 \rho(E_f) \quad (8)$$

The quantized vector potential for a single photon mode with wave vector $\mathbf{k}$ and polarization ϵ is given by

$$\mathbf{A} = \sqrt{\frac{2\pi\hbar c^2}{\omega V}} (\epsilon a e^{i\mathbf{k} \cdot \mathbf{r}} + \epsilon^* a^\dagger e^{-i\mathbf{k} \cdot \mathbf{r}}) \quad (9)$$

where V is the quantization volume and $\omega = c|\mathbf{k}|$.

The initial and final states are tensor products of the charge and the photon states,

$$|i; 0\rangle = |i\rangle_{\text{charge}} \otimes |0\rangle_{\text{photon}} \quad |f; 1\rangle = |f\rangle_{\text{charge}} \otimes |1_{\mathbf{k}, \epsilon}\rangle_{\text{photon}} \quad (10)$$

The operator $\mathbf{A}$ has a and $a^\dagger$ acting on the photon states, and $\epsilon e^{i\mathbf{k}\cdot\mathbf{r}}$ and $\epsilon^* e^{-i\mathbf{k}\cdot\mathbf{r}}$ acting on the charge states. Thus

$$\begin{aligned} \langle f; 1 | H' | i; 0 \rangle &= -\frac{e}{mc} \sqrt{\frac{2\pi\hbar c^2}{\omega V}} \left(\overbrace{\langle 1_{\mathbf{k}, \epsilon} | a | 0 \rangle}^0 \cdot \langle f | \mathbf{p} \cdot \epsilon e^{i\mathbf{k}\cdot\mathbf{r}} | i \rangle + \overbrace{\langle 1_{\mathbf{k}, \epsilon} | a^\dagger | 0 \rangle}^1 \cdot \langle f | \mathbf{p} \cdot \epsilon^* e^{-i\mathbf{k}\cdot\mathbf{r}} | i \rangle \right) \\ &= -\frac{e}{mc} \sqrt{\frac{2\pi\hbar c^2}{\omega V}} \langle f | \mathbf{p} \cdot \epsilon^* e^{-i\mathbf{k}\cdot\mathbf{r}} | i \rangle \end{aligned} \quad (11)$$

In the dipole approximation, we take $e^{-i\mathbf{k}\cdot\mathbf{r}} \approx 1$. With the commutation relation

$$[r_j, H_0] = i\hbar \frac{p_j}{m} \quad (12)$$

we have

$$\begin{aligned} \langle f | \mathbf{p} | i \rangle &= \frac{m}{i\hbar} \langle f | [\mathbf{r}, H_0] | i \rangle \quad |i\rangle, |f\rangle \text{ are eigenstates of } H_0 \\ &= -\frac{m}{i\hbar} \overbrace{(E_f - E_i)}^{-\hbar\omega} \langle f | \mathbf{r} | i \rangle \end{aligned} \quad (13)$$

thus

$$|\langle f; 1 | H' | i; 0 \rangle|^2 = \frac{2\pi e^2 \omega \hbar}{V} |\epsilon^* \cdot \langle f | \mathbf{r} | i \rangle|^2 \quad (14)$$

Consider the big-box quantization with $V = L^3$. The boundary condition requirement for the wavevector is

$$\mathbf{k} = \frac{2\pi}{L} (n_x \hat{\mathbf{x}} + n_y \hat{\mathbf{y}} + n_z \hat{\mathbf{z}}) \quad n_x, n_y, n_z = 0, 1, 2, \dots \quad (15)$$

Thus the number of modes at $\mathbf{k}$ with a given polarization ϵ d^3k is

$$dN(\mathbf{k}, \epsilon) = \left(\frac{L}{2\pi}\right)^3 d^3k = \left(\frac{L}{2\pi}\right)^3 k^2 dk d\Omega = \frac{V \omega^2}{(2\pi)^3 \hbar c^3} dE d\Omega \quad (16)$$

giving the density of final photon states per unit energy per unit solid angle

$$\rho(\hbar\omega, \epsilon) = \frac{dN(\hbar\omega, \epsilon)}{d(\hbar\omega) d\Omega} = \frac{V \omega^2}{(2\pi)^3 \hbar c^3} \quad (17)$$

Finally, plugging this into Fermi's Golden Rule by summing over the two polarizations, we have the transition rate

$$\begin{aligned} \Gamma_{i \rightarrow f} &= \frac{2\pi}{\hbar} \sum_{\epsilon=1,2} \int d\Omega |\langle f; 1 | H' | i; 0 \rangle|^2 \rho(\hbar\omega, \epsilon) \\ &= \frac{e^2 \omega^3}{2\pi \hbar c^3} \sum_{\epsilon=1,2} \int d\Omega |\epsilon^* \cdot \langle f | \mathbf{r} | i \rangle|^2 \quad \text{let } \langle f | \mathbf{r} | i \rangle = |\langle f | \mathbf{r} | i \rangle| \cos \theta \hat{\mathbf{k}} + \sin \theta \cos \phi \epsilon_1 + \sin \theta \sin \phi \epsilon_2 \\ &= \frac{e^2 \omega^3}{2\pi \hbar c^3} |\langle f | \mathbf{r} | i \rangle|^2 \int \sin^2 \theta d\Omega \\ &= \frac{4e^2 \omega^3}{3\hbar c^3} |\langle f | \mathbf{r} | i \rangle|^2 \end{aligned} \quad (18)$$

Specializing for the harmonic oscillator, we have

$$r_j = \sqrt{\frac{\hbar}{2m\omega_0}} (a_j + a_j^\dagger) \quad j = x, y, z \quad (19)$$

For initial state $|i\rangle = |n_x, n_y, n_z\rangle$ with one photon emitted the final state must be $|f\rangle = |n_x - 1, n_y, n_z\rangle$ or $|n_x, n_y - 1, n_z\rangle$ or $|n_x, n_y, n_z - 1\rangle$, giving

$$\langle n_x - 1, n_y, n_z | r_x | n_x, n_y, n_z \rangle = \sqrt{\frac{\hbar}{2m\omega_0}} \sqrt{n_x} \quad (20)$$

similarly for r_y and r_z . Thus the total transition rate is

$$\Gamma_{\text{total}} = \frac{4e^2\omega_0^3}{3\hbar c^3} \cdot \frac{\hbar}{2m\omega_0} (n_x + n_y + n_z) = \frac{2e^2}{3mc^3} \omega_0^2 \cdot n = n\omega_0^2 \tau \quad (21)$$

agreeing with the classical result.

For angular momentum, let the initial and final states be $|i\rangle = |n, l, m\rangle$ and $|f\rangle = |n', l', m'\rangle$. Since operator $\mathbf{r}$ is a linear combination of spherical tensor $T_q^{(1)}$ with $q = -1, 0, 1$, the matrix element $\langle f | \mathbf{r} | i \rangle$ is a linear combinations of $\langle n', l', m' | T_q^{(1)} | n, l, m \rangle$. By Wigner-Eckart theorem,

$$\langle n', l', m' | T_q^{(1)} | n, l, m \rangle = \langle l, m; 1, q | l', m' \rangle \frac{\langle n', l' || T^{(1)} || n, l \rangle}{\sqrt{2l' + 1}} \quad (22)$$

where the first factor is the Clebsch-Gordan coefficient indicating selection rule

$$l - 1 \leq l' \leq l + 1 \quad m' = m + q \quad (23)$$

But $\Delta l = 0$ is removed by parity: under $\mathbf{r} \rightarrow -\mathbf{r}$ the dipole operator is odd: $T_q^{(1)} \rightarrow -T_q^{(1)}$, while $|n, l, m\rangle$ has parity $(-1)^l$, so the matrix element acquires a factor $(-1)^{l'+1+l}$ and vanishes unless $l' + l + 1$ is even, i.e. $l' - l$ is odd (similar argument appears in Jackson section 9.8 on pp436).

This means the emission splits into two channels

$$(n_0, l_0) \rightarrow \begin{cases} (n_0 - 1, l_0 - 1) & \text{with rate } \Gamma_- \\ (n_0 - 1, l_0 + 1) & \text{with rate } \Gamma_+ \end{cases} \quad (24)$$

where the channel rate is given by summing over m', q :

$$\Gamma_{\pm} = \frac{4\omega_0^3 e^2}{3\hbar c^3} \left| \frac{\langle n', l' || T^{(1)} || n, l \rangle}{\sqrt{2l' + 1}} \right|^2 \sum_{m', q} |\langle l, m; 1, q | l', m' \rangle|^2 \quad (25)$$

Without going into the calculation details (which involves specific harmonic oscillator radial solution and summation of Clebsch-Gordan coefficients), we can show that

$$\Gamma_+ = \frac{(l_0 + 1)(n_0 - l_0)}{2l_0 + 1} \Gamma \quad (26)$$

$$\Gamma_- = \frac{l_0(n_0 + l_0 + 1)}{2l_0 + 1} \Gamma \quad (27)$$

Then the mean rate of change of l is

$$\left\langle \frac{dl}{dt} \right\rangle = \Gamma_+ + (-1)\Gamma_- = \left[\frac{n_0 - 2l_0(l_0 + 1)}{2l_0 + 1} \right] \Gamma \rightarrow -n_0 \Gamma \quad \text{as } l_0 \rightarrow n_0 \gg 1 \quad (28)$$

agreeing with the classical result.